

ECONOMICS

HSS 301

Part A (Microeconomics)

Discussion – 1

Market Forces of Demand & Supply

Law of Demand

Ceteris paribus, the quantity demanded of a good falls when the price of the good rises

*Ceteris
paribus*

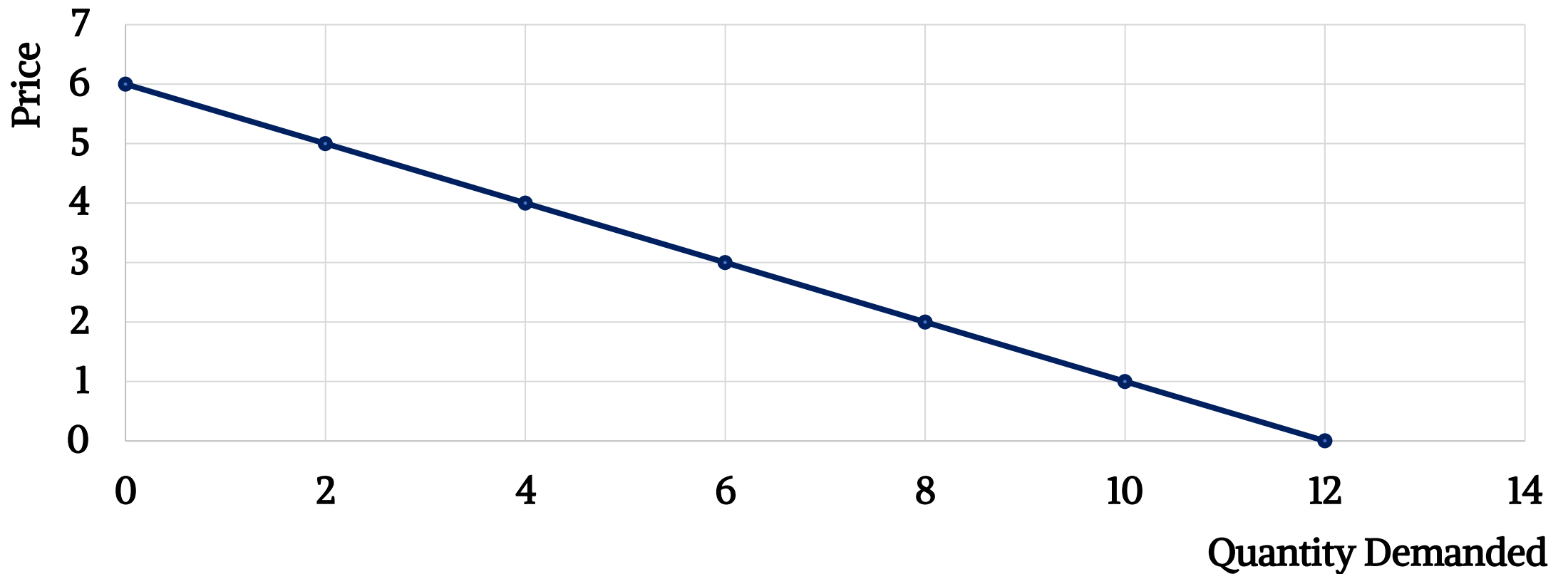
All other things being
equal or unchanged

Demand Schedule

Price (\$) of ice cream cones	Quantity of cones demanded
0	12
1	10
2	8
3	6
4	4
5	2
6	0

Demand Curve

Demand curve for ice cream cones



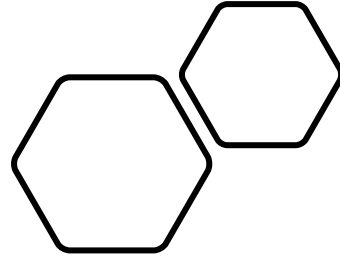
Demand function/equation

$$Q_d = 12 - 2P$$
$$P = 6 - 0.5Q_d$$

P= Price of ice cream cones

Q_d = Quantity of ice cream cones demanded

Meaning of the slope



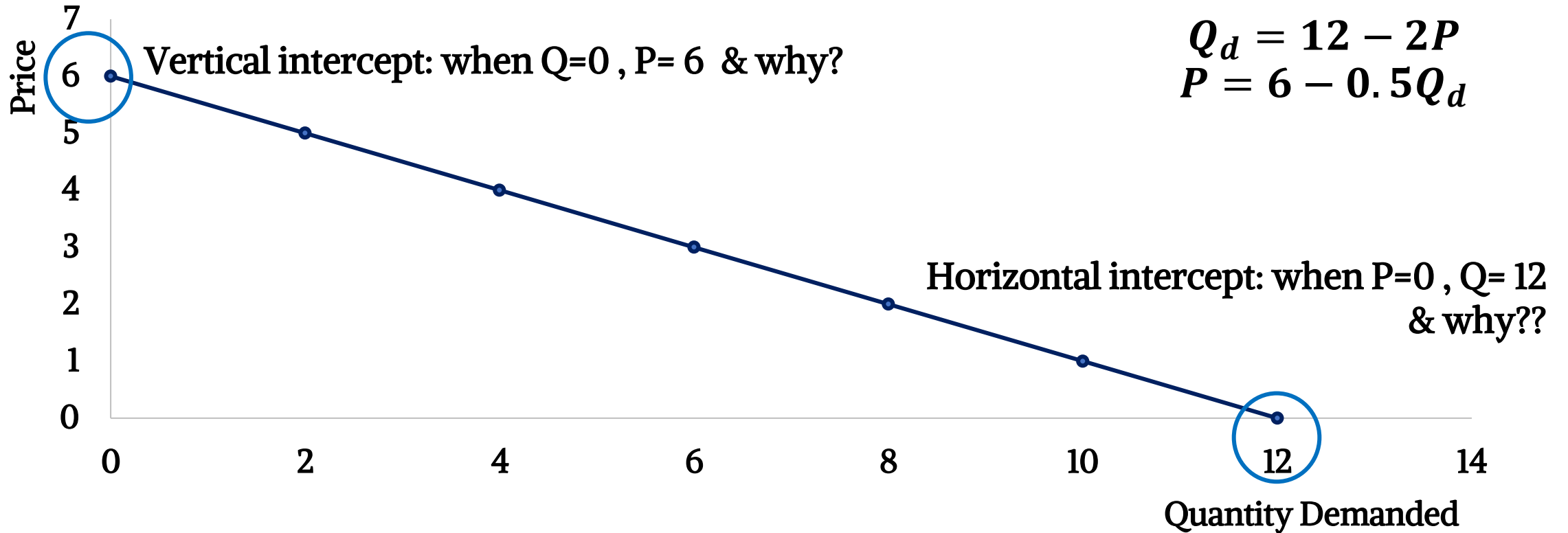
$$Q_d = 12 - 2P$$

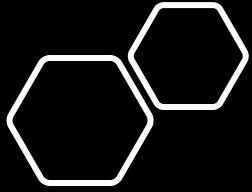
- Here, slope $= \frac{d}{dP} (Q_d) = -2$
- It means: if price rises by 1 unit, quantity of cones demanded falls by 2 units.

Meaning of the intercepts

Demand curve for ice cream cones

$$Q_d = 12 - 2P$$
$$P = 6 - 0.5Q_d$$





Meaning of the intercepts

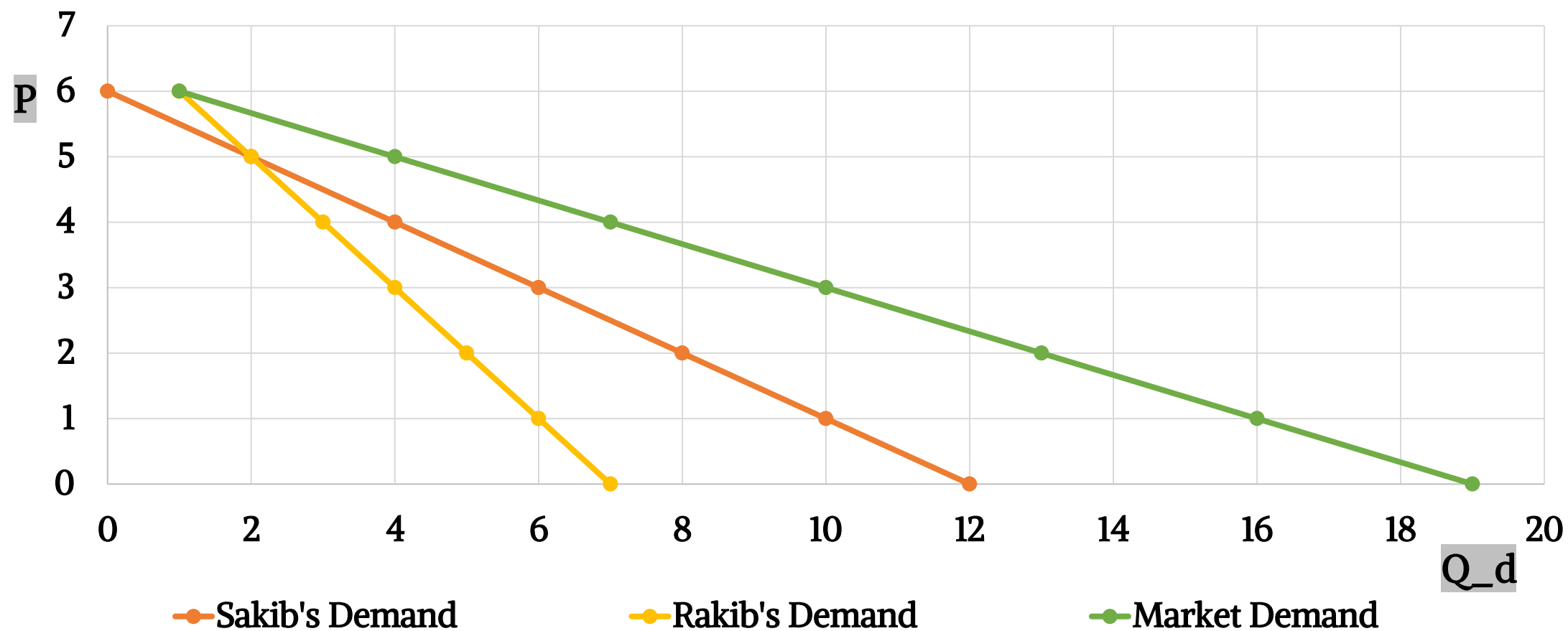
- Vertical intercept shows consumer's reservation price, which is his/her highest willingness to pay for that good
- Horizontal intercept shows the quantity at which consumers total utility is maximum & marginal utility is 0

Individual Demand vs. Market Demand

Individual demand vs. Market demand

Price (\$) of ice cream cones	Quantity of cones demanded by Sakib (Q_{d_1})	Quantity of cones demanded by Rakib (Q_{d_2})	Quantity demanded in the market ($Q_{d_1} + Q_{d_2}$)
0	12	7	19
1	10	6	16
2	8	5	13
3	6	4	10
4	4	3	7
5	2	2	4
6	0	1	1

Individual vs. Market Demand



Individual demand vs. Market demand

-
- Sakib's demand function: $Q_{d_1} = 12 - 2P$
 - Rakib's demand function: $Q_{d_2} = 7 - P$
 - Market demand function: $Q_{d_m} = Q_{d_1} + Q_{d_2}$
so, $Q_{d_m} = 19 - 3P$

Now.....what are those
other things?

Demand Shifters

Income of consumers

Price of related goods

Consumer tastes & preferences

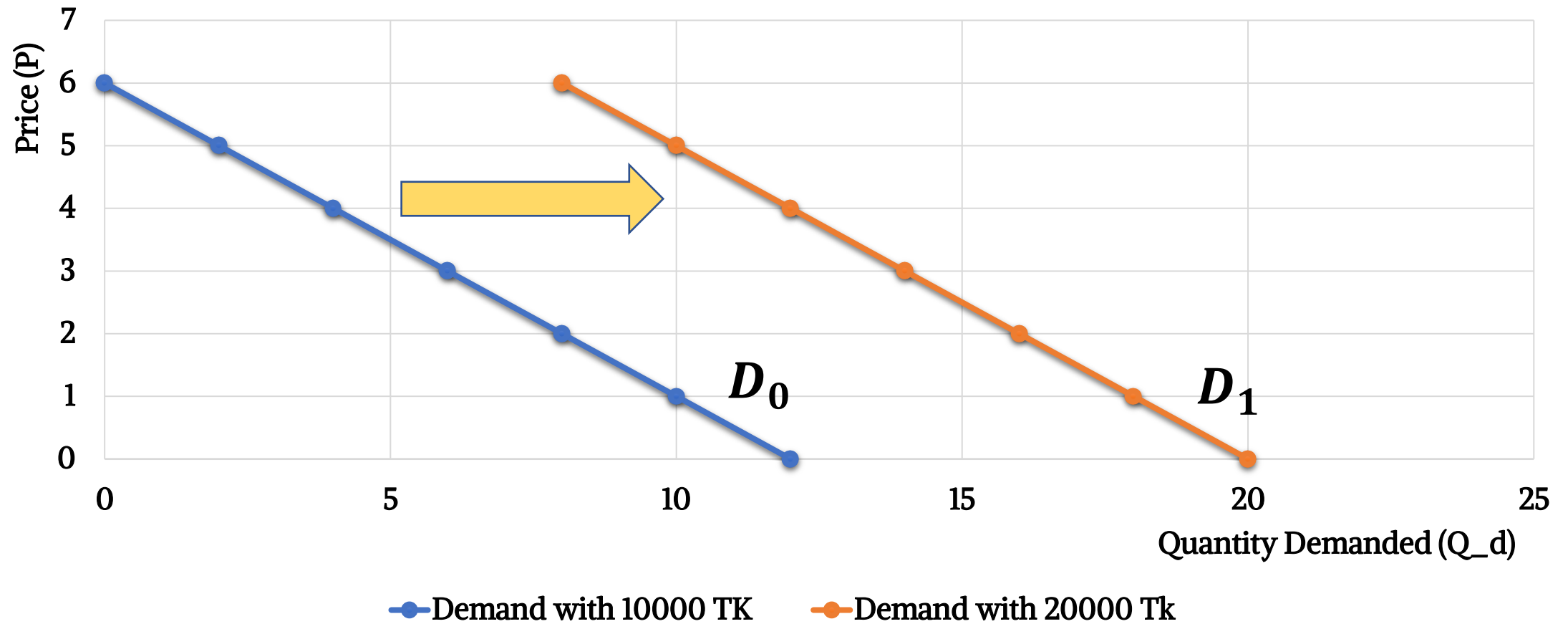
Expectation of consumers

Market size/Number of buyers

Changes in consumer's income

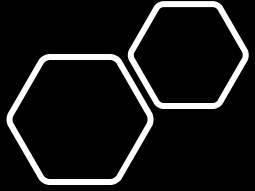
Price (\$) of ice cream cones	Quantity of cones demanded by Sakib with the income of 10000 Tk	Quantity of cones demanded by Sakib with the income of 20000 Tk
0	12	20
1	10	18
2	8	16
3	6	14
4	4	12
5	2	10
6	0	8

Demand Shifts right due to rise in income



But there is a twist

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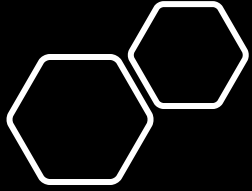
2 types of
goods
regarding
income

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graph LR; A[Normal Good] --- B[Inferior Good];
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A diagram consisting of two dark blue rounded rectangular boxes, one on the left and one on the right. Each box contains white text. A thin, light blue arc connects the top of the left box to the top of the right box, and another similar arc connects the bottom of the left box to the bottom of the right box, forming a frame around the two boxes.

Normal
Good

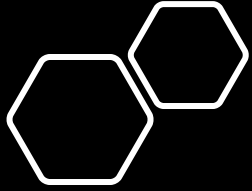
Inferior
Good



Normal Good

A good for which, other things being equal, an increase in income leads to an increase in demand

Example: Coffee, shirt, movie ticket etc.



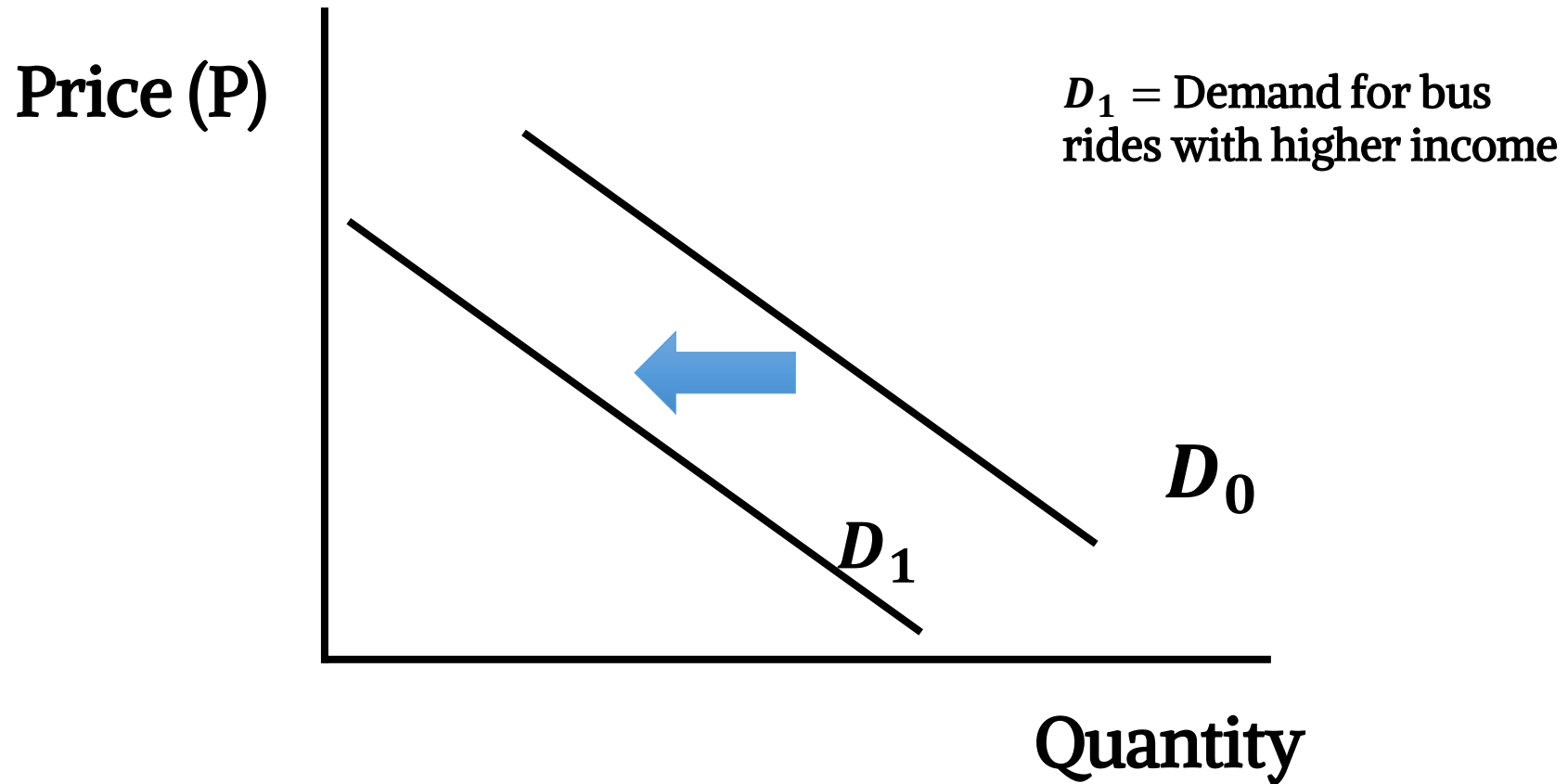
Inferior Good

A good for which, other things being equal, an increase in income leads to a decrease in demand

Demand curve of inferior good shift left when income rises

Example: Bus rides, butter bun, tilapia fish etc.

Demand of Inferior Good when income rises



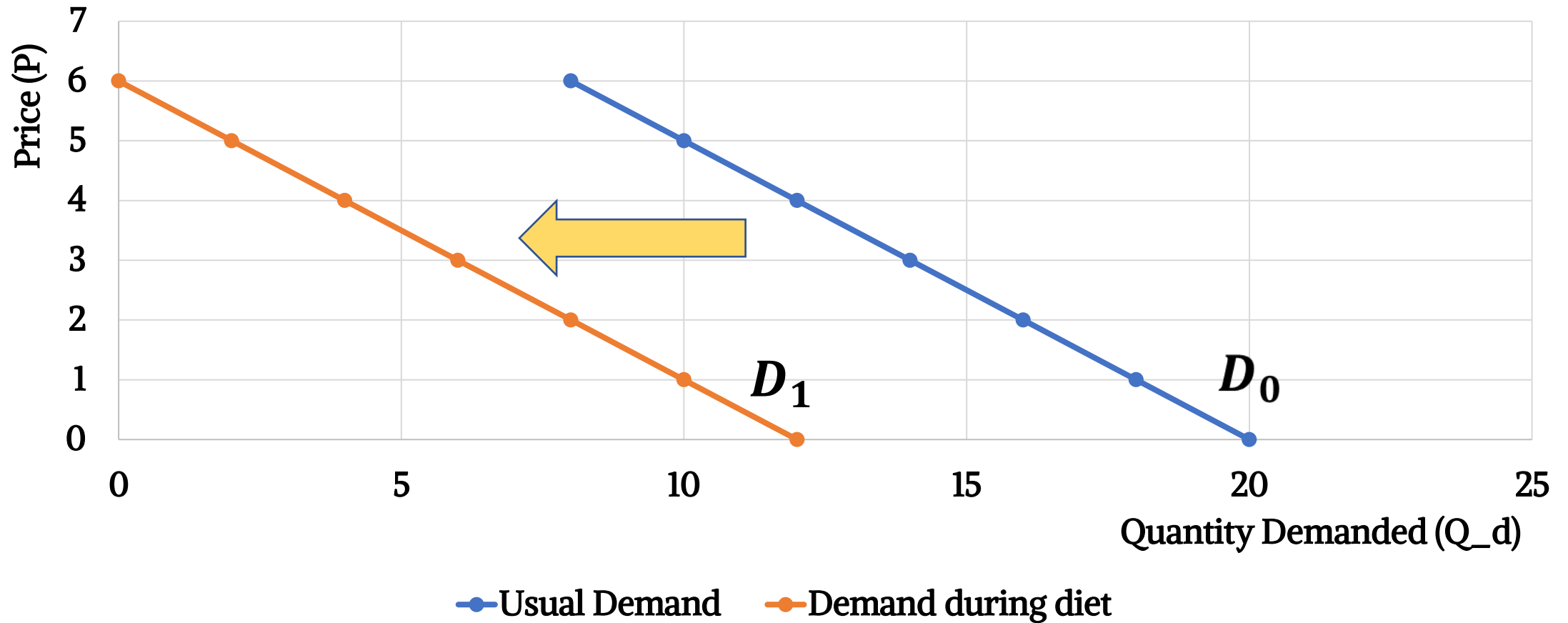
Bonus Question

- If pasta is an inferior good, then the demand curve shifts to the _____ when _____ rises.
- a) Right; the price of pasta
 - b) Right; consumer's income
 - c) Left; the price of pasta
 - d) Left; consumer's income

Changes in consumer's taste/preference

Price (\$) of ice cream cones	Quantity of cones demanded by Fahim usually	Quantity of cones demanded by Fahim when he dislikes it or in a diet
0	20	12
1	18	10
2	16	8
3	14	6
4	12	4
5	10	2
6	8	0

Demand Shifts left due to fall in Taste/Preference



Market Size

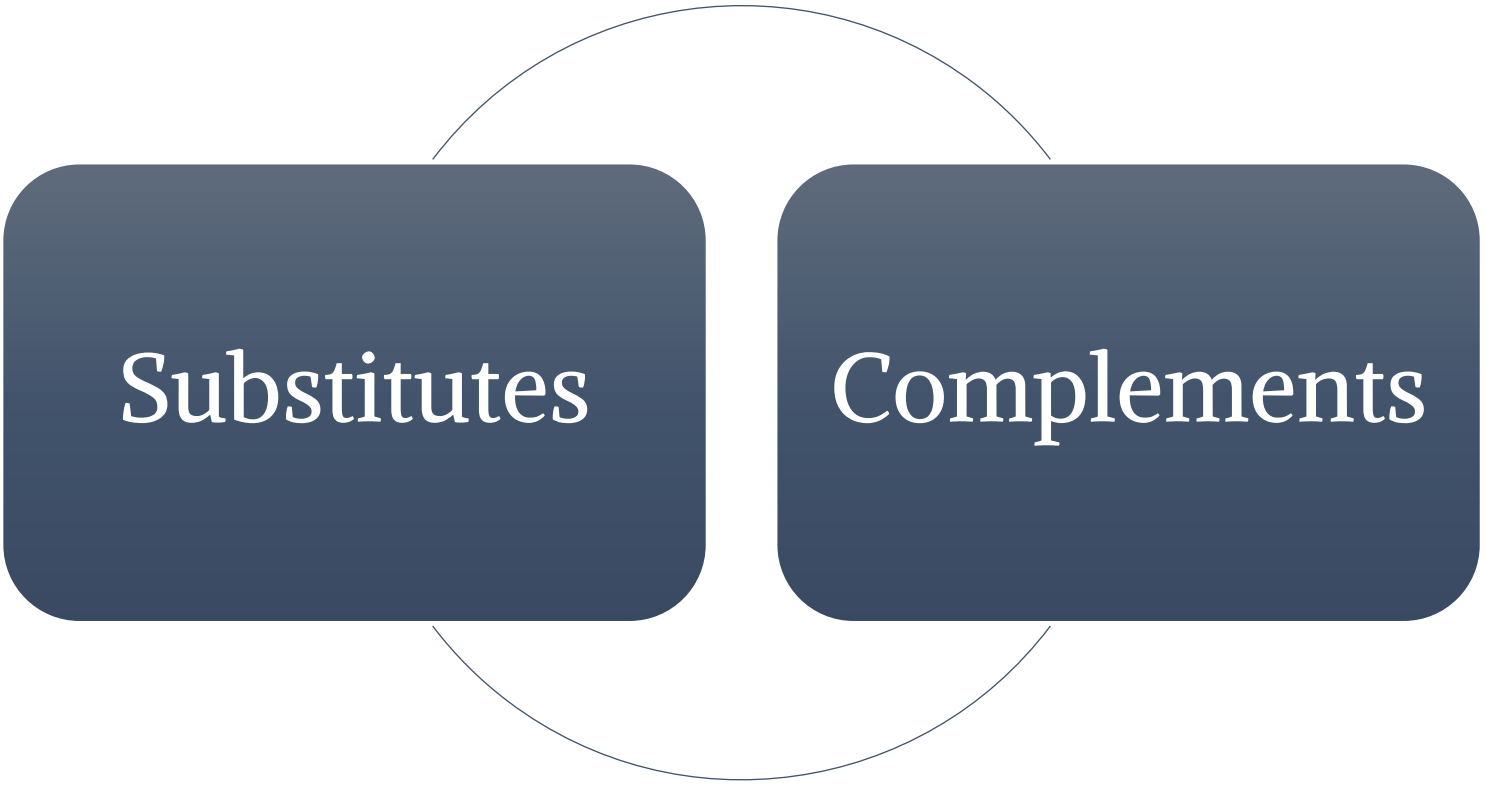
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- When more people want to purchase a product/service, at that time, market demand for that good/service rises & the demand curve shifts to the right.
- When less people want to purchase a product/service, at that time, market demand for that good/service falls & the demand curve shifts to the left.

Consumer Expectations

-
- When consumer expects a higher price in future, his/her demand for that product increases at present & demand curve shifts to the right.
 - When consumer is expecting a lower price in future, his/her demand for that product decreases at present & demand curve shifts to the left.

Price of
related
goods



A diagram consisting of two dark blue rounded rectangular boxes, one on the left and one on the right. The left box contains the word "Substitutes" and the right box contains the word "Complements". Both boxes are white-outlined. A thin, light blue arc connects the top-right corner of the left box to the top-left corner of the right box, and another similar arc connects the bottom-right corner of the left box to the bottom-left corner of the right box, forming a shape like a horizontal oval.

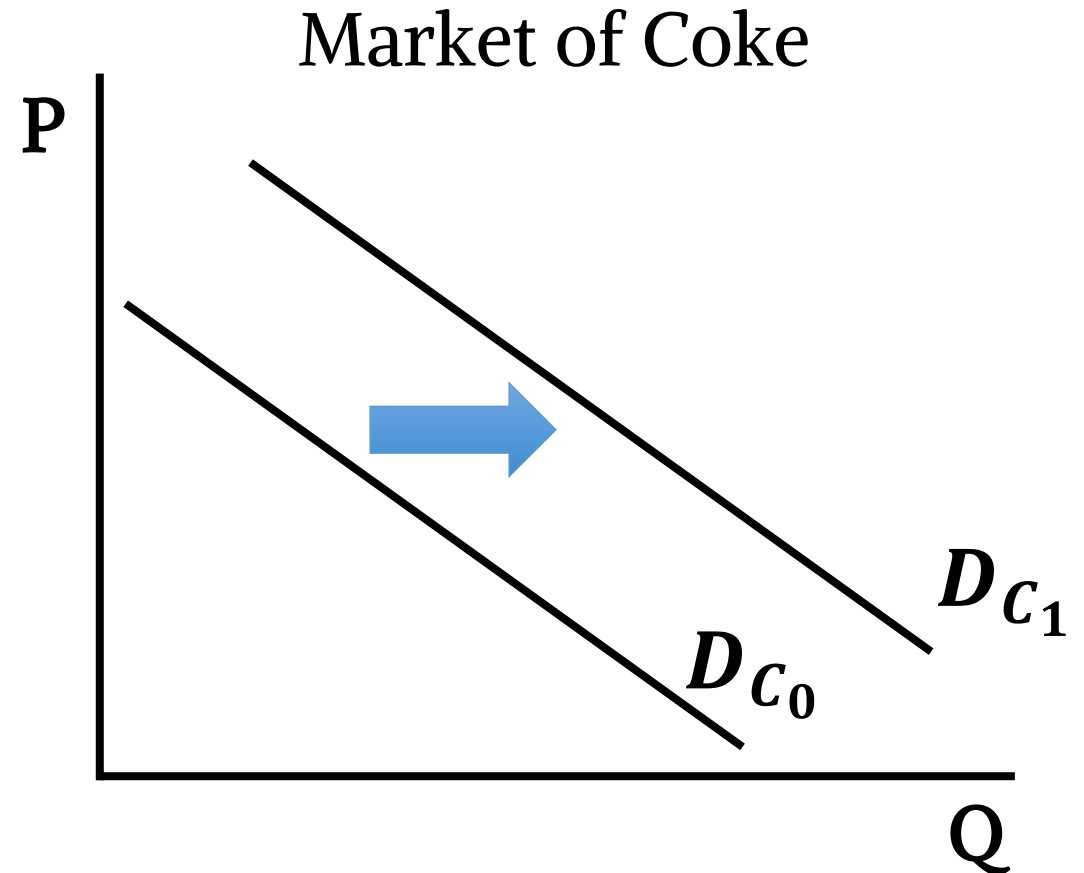
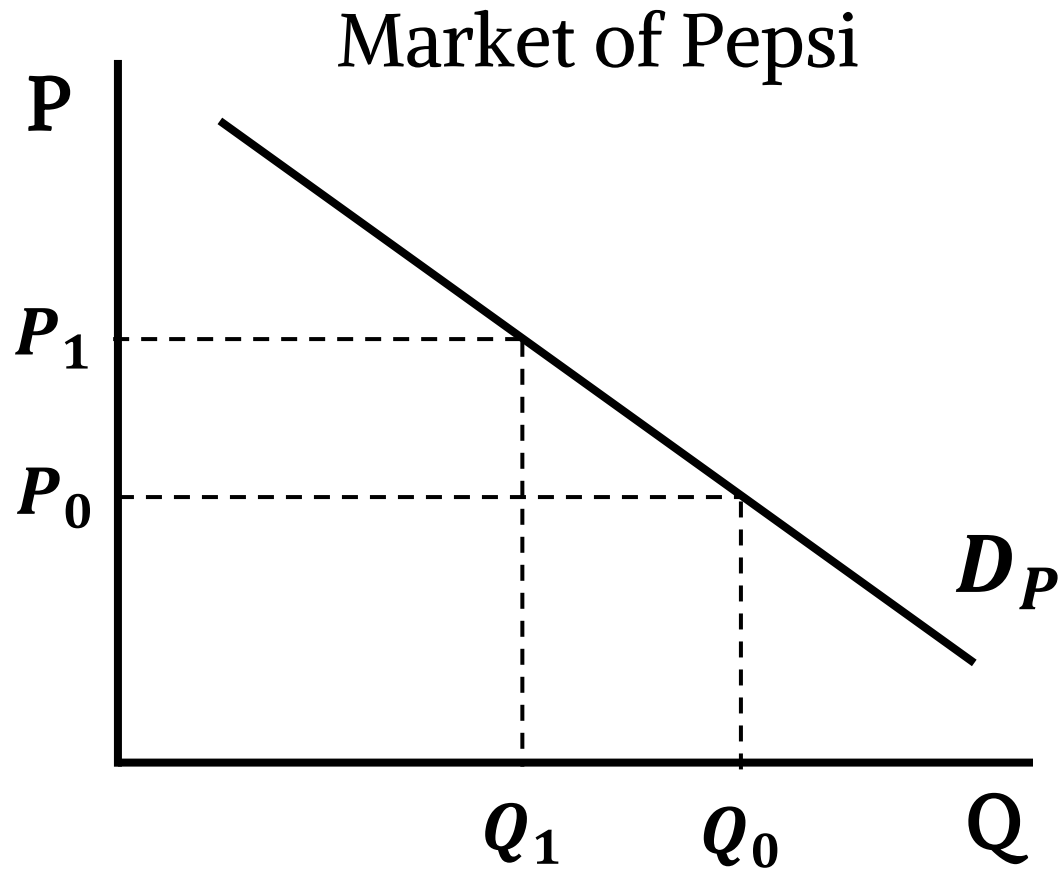
Substitutes

Complements

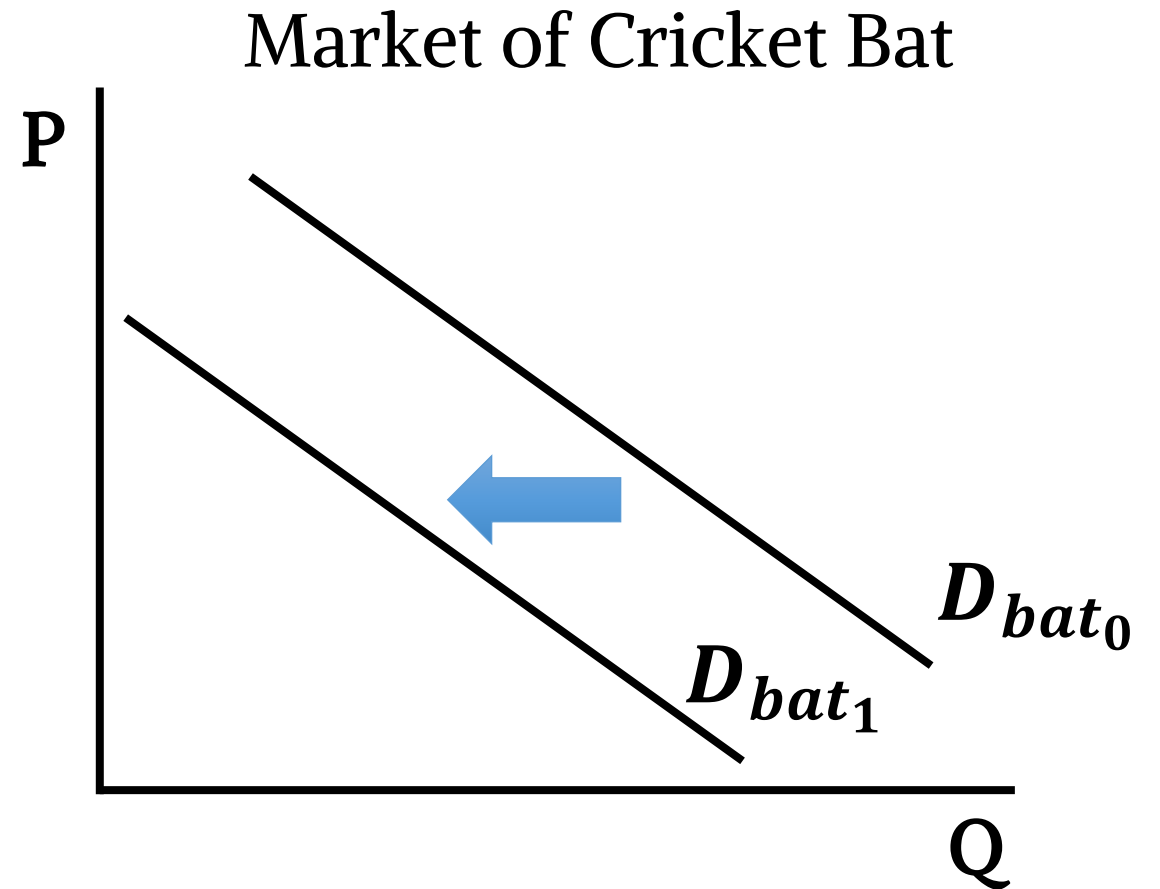
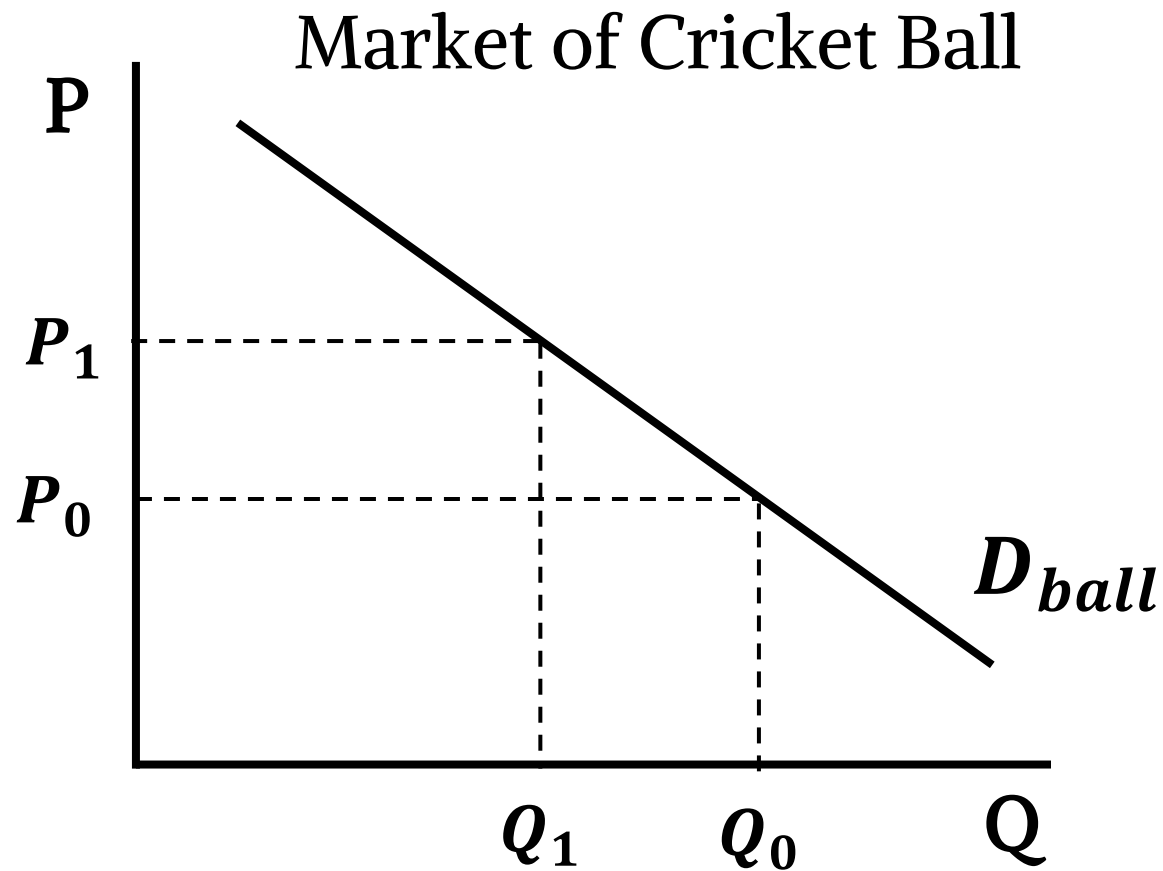
Related goods

- **Substitutes:** two goods are substitutes if the products could be used for the same purpose by the consumers, so that having more of one good causes the consumer to desire less of the other good
- **Complements:** A complementary good is a product or service that provides value to another product or service, so that having less of one good causes the consumer to desire less of the other good

Demand of a product rises (curve shifts right) when price of its substitute rises



Demand of a product falls (curve shifts left) when price of its complement rises





Bonus Question

Which of the following will shift the demand curve for pizza to the right?

- a) An increase in the price of hamburgers, a substitute for pizza
- b) An increase in the price of root beer, a complement to pizza
- c) The departure of college students, as they leave for summer vacation
- d) A decrease in the price of pizza

A decorative graphic on the left side of the slide. It features a large, light blue circle with a white center. Inside the white center is a blue checkmark. To the right of the checkmark are three horizontal blue bars. The entire graphic is set against a background of concentric circles in shades of green and blue.

Bonus Question

- What happens to the demand for a product when the price of the product decreases?
 - a) Demand Stays the Same
 - b) Demand Decreases
 - c) Demand Increases



Movement along
the curve vs. shift
of the curve

Law of Supply

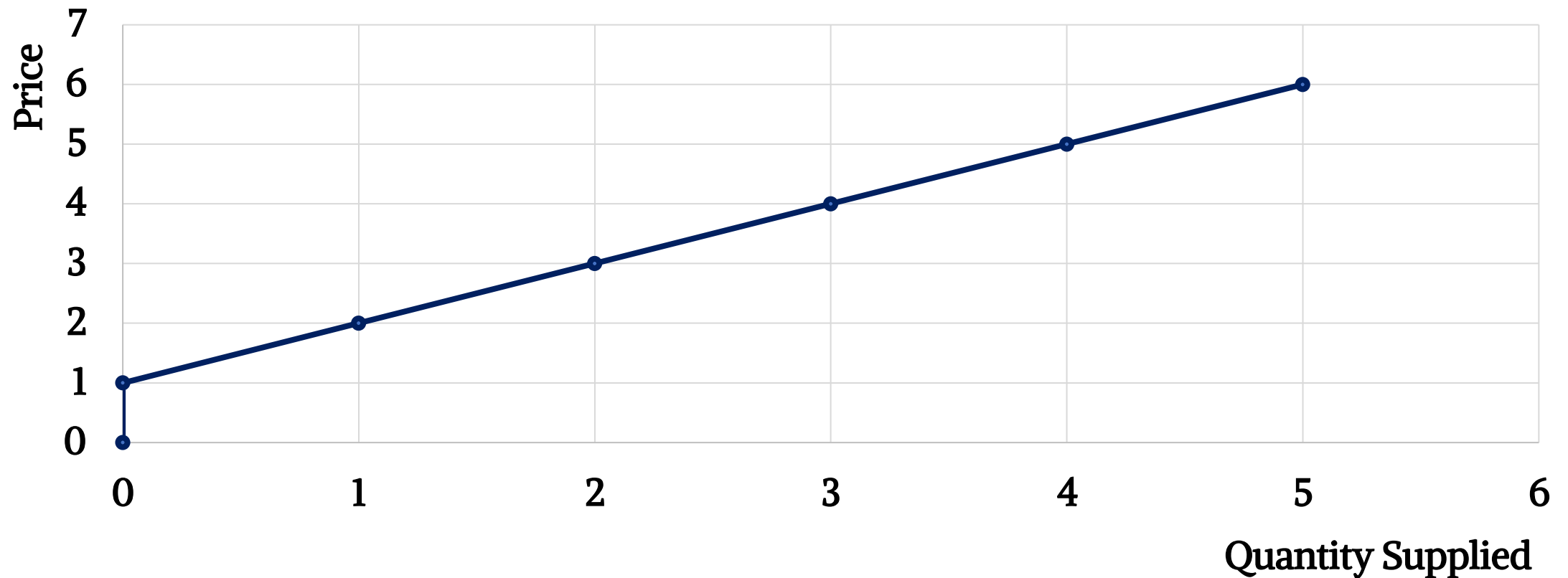
Ceteris paribus (other things being unchanged), the quantity supplied of a good rises when the price of the good rises

Supply Schedule

Price (\$) of ice cream cones	Quantity of cones Supplied
0	0
1	0
2	1
3	2
4	3
5	4
6	5

Supply Curve

Supply curve of ice cream cones



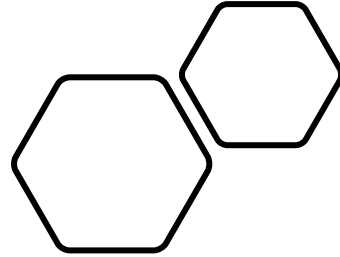
Supply function/equation

$$Q_s = P - 1$$
$$P = 1 + Q_s$$

P= Price of ice cream cones

Q_s = Quantity of ice cream cones supplied

Meaning of the slope



$$Q_s = P - 1$$

- Here, slope $= \frac{d}{dP} (Q_s) = 1$
- It means: if price rises by 1 unit, quantity of cones supplied rises by 1 unit

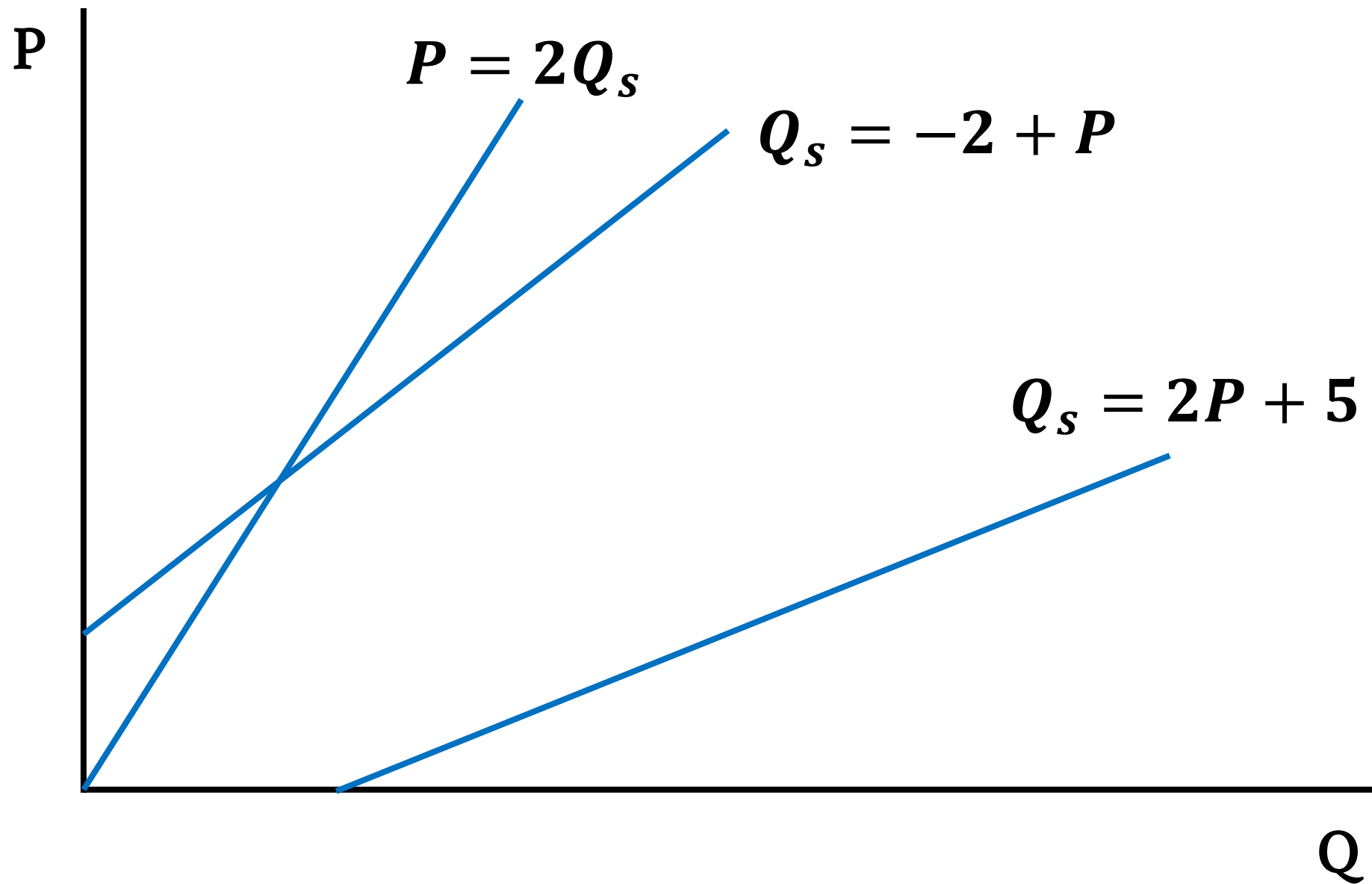
Meaning of intercepts

Firstly, Draw the following
supply curves:

$$Q_s = -2 + P$$

$$Q_s = 2P + 5$$

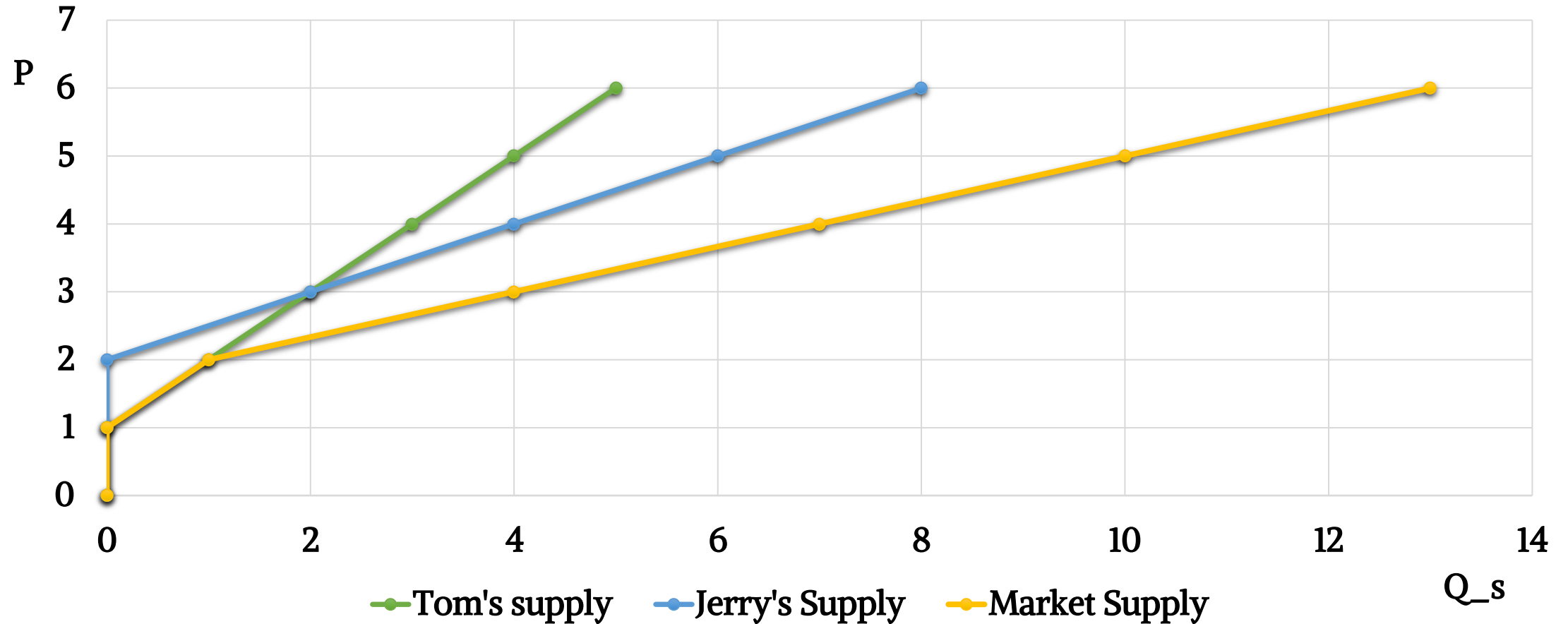
$$P = 2Q_s$$



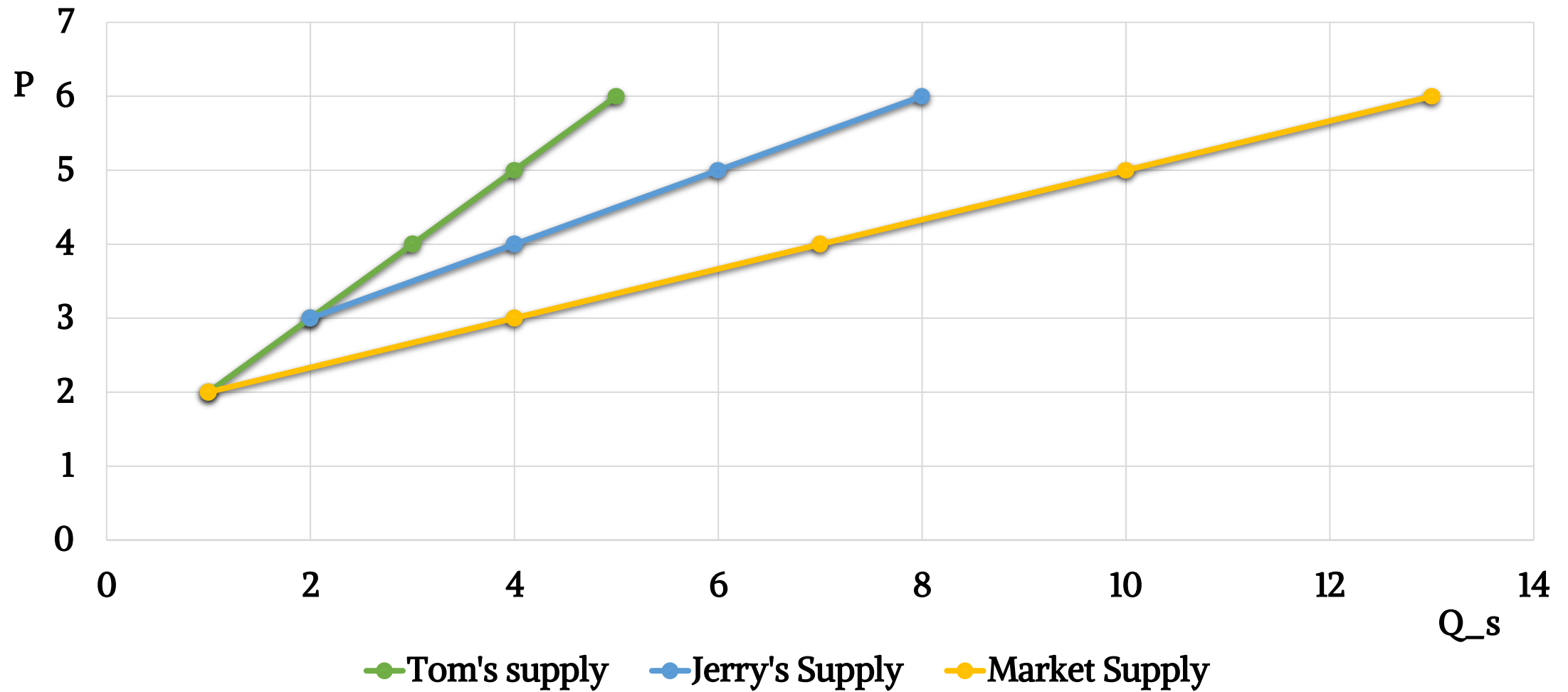
Individual vs. Market Supply

Price (\$) of ice cream cones	Quantity of cones supplied by Tom (Q_{S_1})	Quantity of cones supplied by Jerry (Q_{S_2})	Quantity supplied in the market ($Q_{S_1} + Q_{S_2}$)
0	0	0	0
1	0	0	0
2	1	0	1
3	2	2	4
4	3	4	7
5	4	6	10
6	5	8	13

Individual vs. Market Supply



Without the $Q=0$ values



Individual Supply vs. Market Supply

-
- Tom's supply function: $Q_{S_1} = P - 1$
 - Jerry's supply function: $Q_{S_2} = 2P - 4$
 - Market Supply function: $Q_{S_m} = Q_{S_1} + Q_{S_2}$
so, $Q_{S_m} = 3P - 5$

Supply shifters



Input prices



Production technology



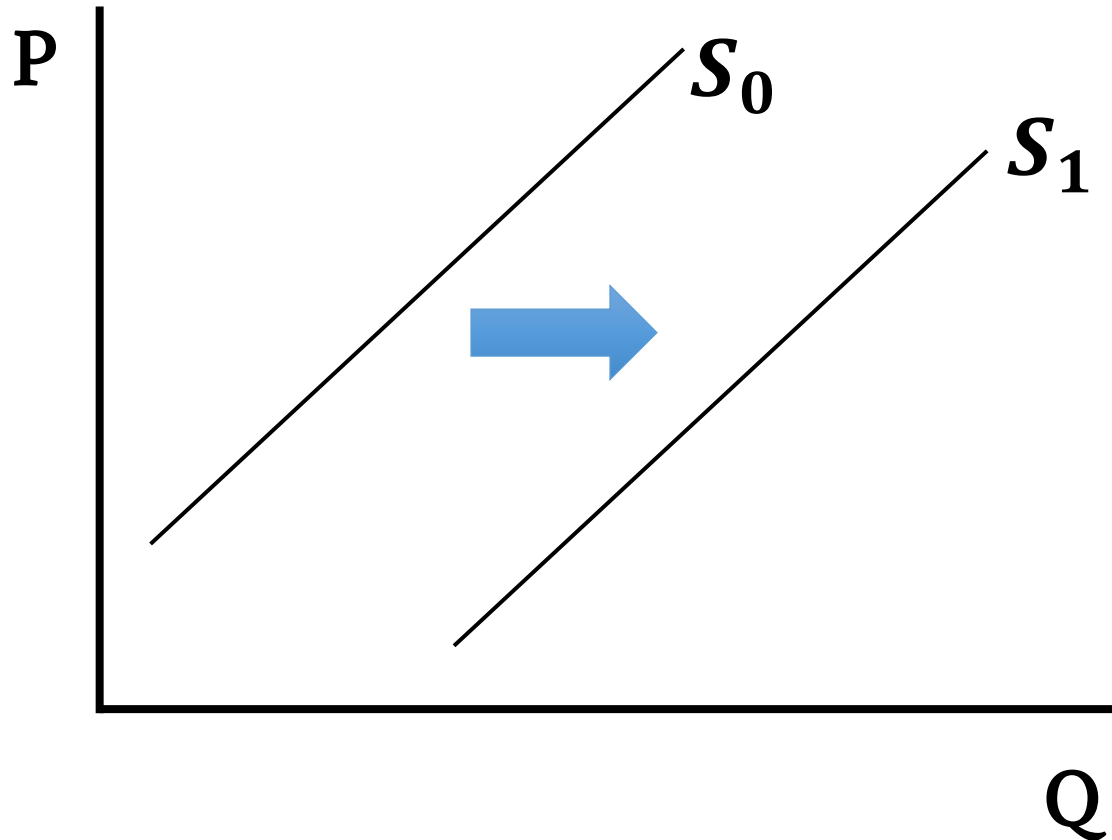
Expectation of sellers



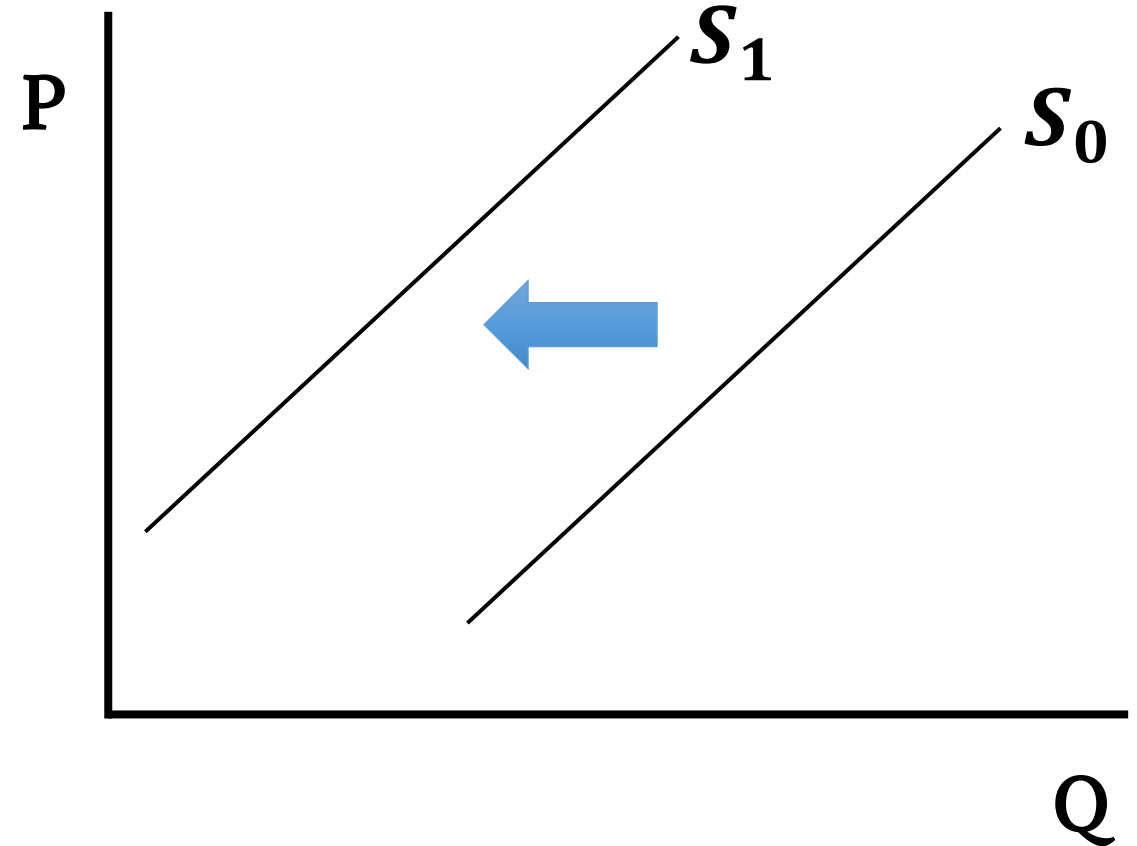
Market size/Number of sellers

Shift of Supply Curve

When supply rises,
curve shifts right



When supply falls,
curve shifts left



Input Prices

When labor/capital/raw material/transport etc. price falls, production cost declines. Supply is increased to gain more profit. Supply curve shifts to the right.

When these sort of input price rises, production cost rises. Supply is decreased to gain more profit or cut back losses. Supply curve shifts to the left.

Production Technology

- When technology improves, productivity rises or production cost declines. Supply is increased. Curve shifts to the right.
- When technology deteriorates, productivity falls or production cost rises. Supply is decreased. Curve shifts to the left.

Seller Expectations

-
- When seller expects a higher price in future, his/her supply for that product decreases at present & supply curve shifts to the left.
 - When seller is expecting a lower price in future, his/her supply for that product increases at present & supply curve shifts to the right.

Market Size/Number of Sellers

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- When more suppliers enter in the market & want to sell a product/service, at that time, market supply for that good/service rises & the supply curve shifts to the right.
- When sellers leave the market of a product/service, at that time, market supply for that good/service falls & the supply curve shifts to the left.

Exercise

Identify whether each of the following trends causes a change in market demand for cable TV. In each case, explain your answer.

- a) The monthly rate paid by cable TV subscribers falls.
- b) The price of satellite TV increases.
- c) Because of growing Internet use, people are choosing to watch less TV.
- d) Cable TV providers choose to make more channels available to their subscribers.

Exercise

Draw a demand curve for music downloads. What happens to it in each of the following scenarios? Why?

- A. The price of iPods falls
- B. The price of music downloads falls
- C. The price of CDs falls

Exercise

Identify whether each of the following trends causes a change in supply or a change in quantity supplied for Pacific salmon.

- a) More sophisticated equipment makes it possible for salmon fishers to increase their catch.
- b) Rising temperatures in the Pacific Ocean radically reduce the salmon stock.
- c) The price of another commonly caught Pacific fish species increases.
- d) The price of Pacific salmon increases.

Exercise

Draw a supply curve for tax return preparation software. What happens to it in each of the following scenarios?

- a) Retailers cut the price of the software.
- b) A technological advance allows the software to be produced at lower cost.
- c) Professional tax return preparers raise the price of the services they provide.

Thank You

